Neural Signal Processing: Grant Proposal

Do changes persist after meditation? Microstate-specific Functional Connectivity on Meditation EEG

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Question→

1) Do meditation-induced changes persist beyond the meditation period (hysteresis), and are they stronger in long-term meditators?

Related works:

Zanesco AP, Skwara AC, King BG, Powers C, Wineberg K, Saron CD (2021) Meditation training modulates brain electric microstates and felt states of awareness. *Hum Brain Mapp* 42:3228–3252. doi:10.1002/hbm.25430

Bréchet L, Michel CM (2022) EEG microstates in altered states of consciousness. *Front Psychol* 13:856697. doi:10.3389/fpsyg.2022.856697

WHY EEG MICROSTATES?

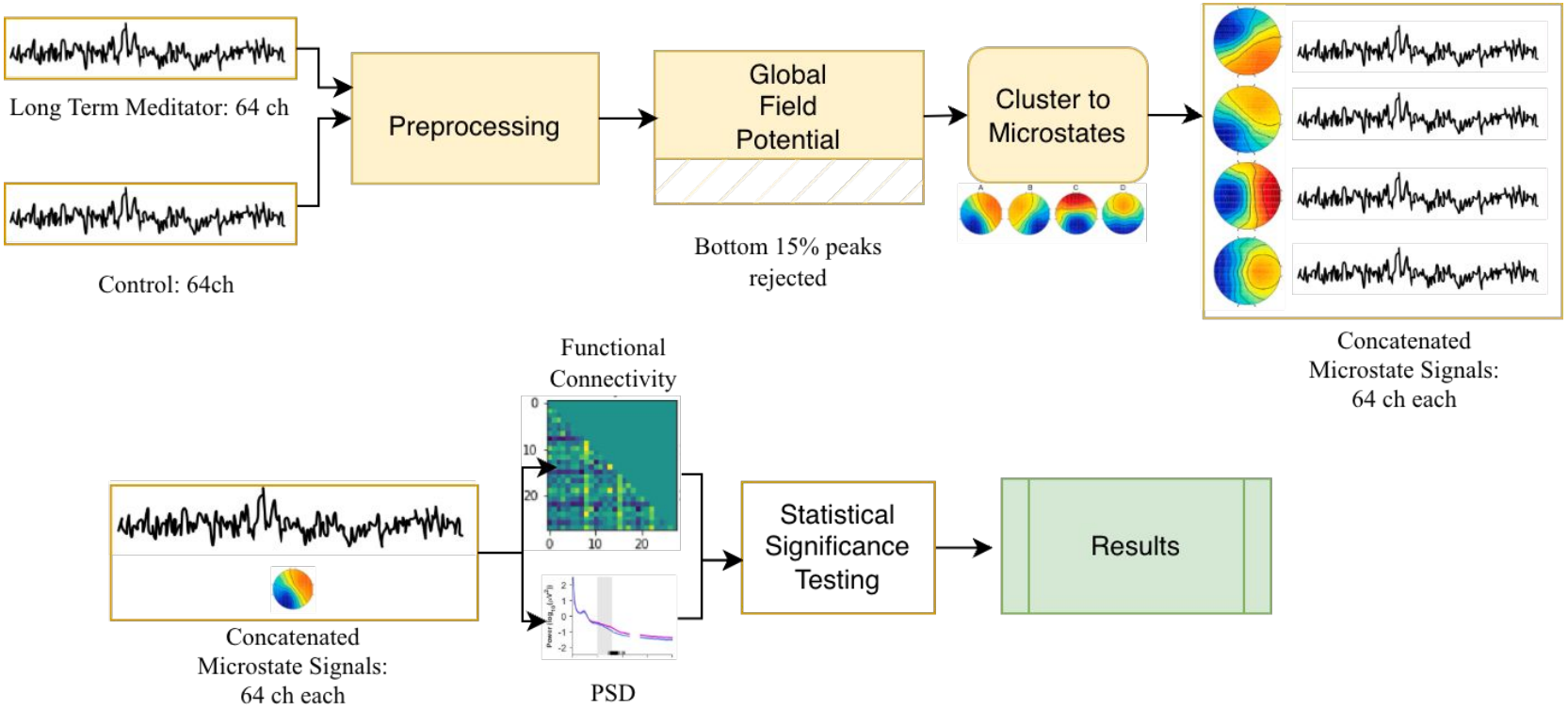
CAPTURING BRAIN DYNAMICS:

- Functional Atoms of Thought
- Beyond Frequency
- Sensitivity to states

ADVANTAGES OF THIS STUDY:

- Hysteresis Detection
- Connectivity mapping
- Noise Reduction

Expected Workflow



Pipeline based on: Hao Z, Zhai X, Cheng D, Pan Y, Dou W (2022) EEG microstate-specific functional connectivity and stroke-related alterations in brain dynamics. *Front Neurosci* 16:848737. doi:10.3389/fnins.2022.848737